

A1 production kernel manifest – verifier calls the REAL N-001 PDE backend (v1.3)

TECT verification-first repository · claim A1-PRODUCTION-KERNEL-MANIFEST

first issued 2026-06-23 · this version issued 2026-06-23 · v1.3

1. Purpose and scope

A production config is the numerical implementation of A2/A3 only if (i) its stored parameters are mutually consistent, (ii) it lies in the analytic branch, and (iii) it is the kernel actually implemented by the canonical N-001 microscopic PDE solver. v1.2 established (i)+(ii) but checked (iii) against the WRONG backend (Math424, a variational-Hartree readings comparison). v1.3 closes (iii) by calling the REAL N-001 solver. Verification: [codes/foundations/a1_kernel_checks.py](#) v1.3 (11/11 PASS); backend [codes/foundations/actual_n001_pde_backend.py](#); config [claims/A1-PRODUCTION-KERNEL-MANIFEST/canonical_n001_kernel.json](#).

2. The real N-001 backend (v1.3)

The canonical N-001 solver fixes the kinetic coefficients via

$$(r, Z) = \text{kinetic_coefficients}(\text{mu2_shell}, Y, q_0) : \quad r = \text{mu2_shell} + Y q_0^4, \quad Z = -2Y q_0^2, \quad (1)$$

(`Codes/pde/continuation_mu2_v25.py`) and realises the linear operator as the scalar Bloch symbol

$$D_{\text{backend}}(k) = r + Z s_2 + Y s_2^2, \quad s_2 = |k|^2 \quad (2)$$

(`bloch_matrix_linear` in `Codes/pde/bloch_linearization.py`, residual $r\Psi - Z\nabla^2\Psi + Y\nabla^4\Psi$). Both are vendored VERBATIM into `actual_n001_pde_backend.py` with the originals pinned by sha256. The verifier (a) confirms `kinetic_coefficients(mu2_shell, Y, q0) = (r_zero, Z)` to 10^{-12} — the stored fields ARE the solver’s own coefficients — and (b) confirms $D_{\text{backend}}(k) = \text{r_zero} + Z|k|^2 + Y|k|^4$ to 8.9×10^{-16} .

3. Faithfulness, provenance, and teeth

When the frozen TECT2 solver tree is reachable, the verifier imports the ORIGINAL `continuation_mu2_v25.py` and asserts the vendored `kinetic_coefficients` reproduces it exactly (0 over a random sweep); otherwise it reports the cross-check SKIPPED and relies on the pinned sha256 (`continuation 6db0393a..`, `bloch 337bd6de..`). The reader has teeth: corrupting the stored q_0 by 1% makes `kinetic_coefficients` diverge from the stored (r, Z) by $O(10^{-2})$.

4. Status: gates pass, certification pending

The canonical N-001 config (`r_zero = 0.21903`, `mu2_shell = 0.005`, `q0 = 0.6801748`, `Z = -0.92528`, `Y = 1`) clears both manifests ($\delta_Z = \delta_r = 0$, $\delta_m = 4.3 \times 10^{-18}$, `mu2_shell > 0`) AND matches the real N-001 backend. The script reports “numerical gates pass; operator certification pending”: the card stays T1 OPEN pending the operator’s independent run of the three-file bundle and certification act.

5. Devil’s-advocate

(α) “*Vendored copy is not the real backend.*” VALID-with-mitigation — the vendored functions are byte-faithful to the originals (verified when reachable; sha256-pinned otherwise), so the bundle is self-contained without misrepresenting the source. (β) “ *$s_2 = |k|^2$ ignores the anisotropic $M_{\text{family}}/M_{\text{lock}}/M_{\text{shellbias}}$ terms.*” VALID — v1.3 certifies the SCALAR linear symbol only; the anisotropic corrections are a separate operator-curvature question (mainline production-Hessian), not the A1 scalar kernel. (γ) “*Gates pass = certified.*” DISMISSED — the script and this note both state certification is pending; T1 OPEN retained. Quantitative sanity check: backend symbol residual 8.9×10^{-16} ; corrupted- q_0 divergence $\sim 9 \times 10^{-3}$ in r .

6. Result footer

Result ID:	A1-PRODUCTION-KERNEL-MANIFEST (OPEN, v1.3)
Precise statement:	Verifier reads 5 stored fields AS-IS, calls the REAL N-001 backend (kinetic_coefficients + bloch scalar symbol, vendored from continuation_mu2_v25.py + bloch_linearization.py): kinetic_coefficients(mu2_shell, Y,q0)=(r_zero,Z) and D_backend(k)=r_zero+Z k^2+Y k^4 (residual 8.9e-16); faithfulness vs original + sha256 pins; delta_{Z,m,r}<=eps; mu2_shell>0. Scalar symbol only.
Scope:	N-001 scalar kernel certification; CONSISTENCY + BRANCH.
Dependencies:	A1-KERNEL-IDENTITY; A1-SCALAR-ANALYTIC-BRANCH.
Bundle:	canonical_n001_kernel.json + actual_n001_pde_backend.py + a1_kernel_checks.py (travel together).
Evidence grade:	ANALYTIC + EXECUTED (11/11) -- pending operator independent run.
Reproduction command:	python codes/foundations/a1_kernel_checks.py
Expected output:	11/11 PASS; kinetic_coefficients match stored; backend symbol matches; faithfulness verified or SKIPPED.
Tier:	T1 OPEN (gates pass; certification pending).
Falsifier:	a backend-source change (hash mismatch), a stored-field edit breaking a delta, or kinetic_coefficients not reproducing the stored (r_zero,Z) refuses use.
No-overclaim:	scalar symbol only (anisotropic M-terms excluded); gates passing != certification; canonical config NOT yet operator-certified, NOT yet A2/A3-citable.
Next required action:	Operator independently runs the bundle (TECT2/Contents reachable) and certifies the N-001 ANALYTIC-BRANCH manifest.